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Executive Summary

I have spent the last 7+ years building financial platforms that handle billions in daily volume while staying live through market volatility, regulatory shifts, and 24/7 global operations. Prior to that, 12 years delivering high-stakes enterprise systems across other domains at scale. All 19 years: production code, real consequences, systems that can't go down.

I have moved from code to architecture to platform leadership across global markets, working inside the constraints that matter: regulatory compliance, sub-millisecond latency, real money at stake. Lately, I have been taking that same engineering discipline into AI -- building governance frameworks, multi-agent systems, and ML audit tools designed the way regulated environments demand them.

What I bring: the judgment to build platforms that last, the technical range to solve problems from data layer to deployment pipeline, and enough scar tissue to know what breaks systems at scale.

Technical Expertise

- **Full-Stack Engineering:** Led core development of backend systems using Java/Spring Boot and Python, while also delivering frontend solutions when needed. Built containerized, cloud-native platforms on AWS, orchestrated by Kubernetes and Docker serving clients in major financial markets.
 - **Event-Driven Architectures:** Developed real-time data feeds and trade workflows with Kafka, Solace, GraphQL, and REST APIs, supporting \$1B+ daily transaction volume.
 - **Performance and Uptime:** Delivered sub-millisecond market data and trade execution using FIX protocol, multicast, and KDB+ optimization. Improved core platform latency up to 15% and throughput by 10%.
 - **DevSecOps & Compliance:** Implemented CI/CD pipelines using secure coding standards (OWASP). Built in audit and regulatory compliance (MiFID II, Dodd-Frank) from the start, eliminating audit findings and reducing security issues by 40%. Deployments routinely passed all reviews.
 - **Data Engineering:** Built scalable data layers with PostgreSQL, Oracle, MongoDB, enabling fast, reliable analytics and compliance reporting across jurisdictions.
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Applied Artificial Intelligence (AI)

Built working systems across the control, evaluation, and defense layers of production AI.

- Constitutional governance SDK that evaluates every LLM output against versioned plain-English rules with an append-only audit log, and a pre-execution code validator that blocks SQL injection, unsafe commands, and secrets exposure via AST rewriting before generated code reaches a runtime.
- Full-stack fairness audit platform calculating demographic parity and disparate impact across protected groups for financial ML models, and a prompt injection defense system with four layered strategies and live effectiveness tracking across multiple LLM backends.
- Multi-agent code reviewer that runs security, performance, and quality checks against git diffs simultaneously, and a real-time trade risk monitor on Kafka with dead letter queue handling.

Research Outputs / Preprints: SSRN (DOI): <https://dx.doi.org/10.2139/ssrn.6384060>.

Portfolio Repository: <https://github.com/sushilduseja>

Professional Experience

Barclays | Vice President, Platform Modernization | Singapore March 2023 – March 2025

Engineering Impact & Business Value

- **Enterprise Platform Modernization:** Led code and delivery for FX modernization: rebuilt core trading systems as microservices and cloud workloads, serving global clients and achieving 99.97% uptime.
- **Scalable Trading Infrastructure Design:** Built out infrastructure for 100+ institutions across APAC and EMEA, handling \$1B+ daily volume with full regulatory compliance.
- **Real-Time Data & AI Analytics Enablement:** Designed and deployed real-time analytics pipelines, enabling machine-learning pilots and consistent operations under regulatory audits.
- **Embedded DevSecOps & Regulatory Compliance:** Integrated DevSecOps from project start. Automated vulnerability checks into release lifecycle; ensured platforms met MiFID II and Dodd-Frank standards.

Innovation & Operational Excellence

- **Automation Programs:** Delivered and validated new pricing and workflow modules in direct client pilots, shifting manual steps to code and reducing intervention by 40%.
- **Metrics & Reliability:** Developed and maintained reliability monitoring tools -- delivered multi-cycle, incident-free operation.

Commerzbank | Vice President, Algo Trading Solutions | Singapore September 2017 – July 2022

Platform Engineering & Delivery

- **Global Team Technical Leadership:** Shipped scalable Java trading platforms for APAC and EMEA markets; led global team through multi-market releases supporting \$500M+ daily trades.
- **Market Expansion & Product Launch:** Delivered four successful platform upgrades; drove backend reliability, integration, and support for new market launches (e.g., Japan TFX, \$10M+ in new annual revenue).

System Optimization & Compliance

- **Market Data & Latency Improvement:** Improved market data processing and trade execution: engineered FIX bridges and custom adapters for 100K+ orders per day at sub-millisecond latency.
- **Real-Time Risk Management:** Built real-time risk modules, raising P&L accuracy by 25%. Platforms routinely passed strict global compliance checks.

IRIS Software Inc. | Associate Project Manager | India | 2017 – 2017

- **Client:** Bank of Montreal - DTCC trade confirmation and settlement platform
- Led requirements, analysis, and implementation of platform enhancements for high-volume trade processing
- Reduced application memory consumption by 30% through GC log analysis and systematic code refactoring

Mahindra Comviva | Technical Manager | India | 2016 – 2017

- **Product:** PreTUPS - prepaid recharge distribution and mobile account management platform
- Directed product development, release management, and feature delivery across distributed teams
- Cut AWS cloud spending by 25% through strategic optimization and best practice implementation
- Improved application stability by 15% by identifying and resolving code inefficiencies

R Systems International | Associate Technical Architect | India | US | 2015 – 2016

- **Client:** DMA Inc. Tax Interchange for Retail - enterprise point-of-sale tax solution
- Rebuilt client relationship and trust through consistent delivery and transparent communication
- Established architectural standards, design patterns, and code review frameworks
- Introduced Test-Driven Development (TDD), achieving 40% reduction in production defects
- Diagnosed and resolved performance bottlenecks using JVisualVM profiling tools

JK Tech | Project Lead | India | UK | 2009 – 2015

- **Client:** Specsavers Optical Group - enterprise retail and supply chain systems
- Managed complete project lifecycles from initiation through post-implementation review
- Improved Agile delivery efficiency by 20% through process refinement and team coaching
- Promoted three times in six years: System Analyst → Technical Lead → Project Lead
- Recognized with "Star of the Iteration" awards (2012, 2013) and "Star of the Quarter" (June 2011)

Fiserv | Senior Software Engineer | India | US | 2007 – 2009

- **Client:** Fiserv Trade Order Management - institutional trading platform
- Designed and built automated test framework executing 500+ tests in 7 minutes, saving 40% processing time
- Developed proof-of-concept solutions using emerging technologies (OpenESB, Apache ServiceMix)
- Delivered specifications, program changes, and comprehensive technical documentation

JK Tech | Software Engineer | India | UK | 2005 – 2007

- **Internal Products:** iTimer timesheet application and corporate website modernization
- Introduced automated unit testing frameworks, reducing manual testing effort by one man-day monthly
- Mastered Agile and iterative development methodologies in production environments
- Supported QA, UAT, and production stabilization across multiple concurrent projects

Certifications & Executive Education

- **Technology Leadership:** AGI Strategy & Future of AI (2026), Cloud-Native & Multicloud Applications (2022), Blockchain Essentials (2022), Foundations of Cybersecurity (2023)
- **Leadership & Governance:** ISO/IEC 27001:2022 Lead Auditor (2025), AI Security & Governance (2025), Cybersecurity Governance Risk & Compliance (2024)
- **Executive Development:** Generative AI for Executives, IBM (2024), Professional Scrum Master (PSM I) (2020)

Key Differentiators

- **Proven Reliability & Scale:** Build and maintain high-volume platforms with measurable uptime and compliance, supporting real-world demands in global markets.
- **Global Platform Execution:** Ship and operate software for APAC, EMEA and AMER, meeting strict regulatory standards and integration needs.
- **Practical Innovation:** Implement new technologies and DevSecOps practices hands-on to strengthen platform resilience and future growth.